

01 PROBABILITY FOUNDATIONS (lecture 2, notes p012-p050)

1. What statistics is (p012-p016) - not examined, but the framing

statistics = the science of COLLECTION, PRESENTATION, ANALYSIS, INTERPRETATION of numerical data

"All statistics are numerical statements, but not all numerical statements are statistics."

Ram has 100 Rs in his pocket -> NOT statistics (one fact, no data set)

avg age of 5th class is 10 yrs -> IS statistics (a derived numerical statement)

Four limitations the slides list (worth a one-line recall, low exam weight): 1. not suited to qualitative phenomena 2. does not study individuals 3. statistical laws are not exact 4. liable to be misused (the customer-satisfaction survey only of loyal customers)

2. Probability, the intuition (p018-p019)

P near 1	= event extremely likely
P = 0.5	= a coin-flip situation
P near 0	= rare event
$0 \leq P(E) \leq 1$	ALWAYS

0 ————— 0.5 ————— 1
 impossible even certain
 | | |
 rare coin flip common

3. Random experiment (p023)

DEFINITION

an experiment whose outcome cannot be predicted with certainty in advance, although ALL possible outcomes are known

EXPERIMENT | POSSIBLE OUTCOMES

toss a coin | H, T

roll a die | 1,2,3,4,5,6

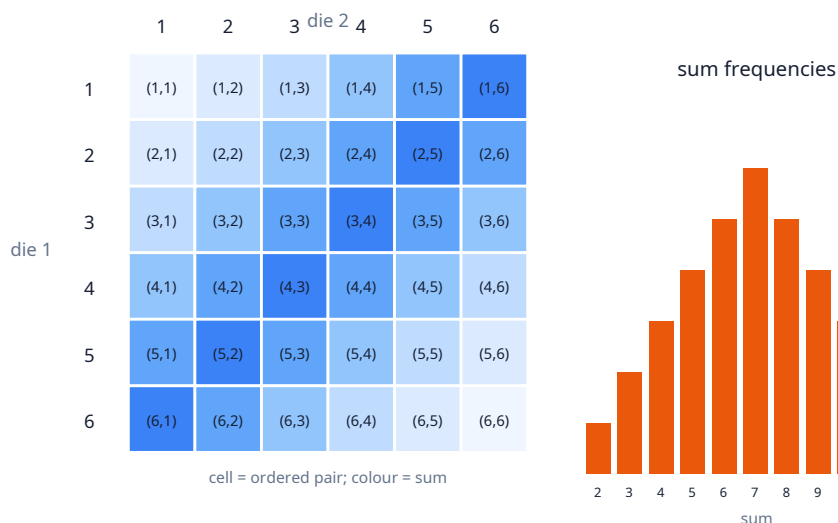
draw a card | any one of 52

pick a student | any student in the class

note: we KNOW all outcomes, we do not know WHICH ONE occurs

4. Sample space S (p024)

Two dice: 36 ordered outcomes



The 36 ordered outcomes for two dice

S = set of ALL possible outcomes		
one coin	$S = \{H, T\}$	2 outcomes
one die	$S = \{1, 2, 3, 4, 5, 6\}$	6 outcomes
two coins	$S = \{HH, HT, TH, TT\}$	4 outcomes
two dice	$S = \{(1,1), (1,2), \dots, (6,6)\}$	$6 \times 6 = 36$ outcomes

THE COUNTING RULE THAT SAVES MARKS

two dice are counted as 36 ORDERED pairs, not 21 unordered

(1,2) and (2,1) are DIFFERENT outcomes

every dice probability (1/36, 2/36, 3/36...) assumes 36

The 36 ordered-pair positions, with each cell displaying its sum:

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1 2 3 4 5 6 <- die 2
1 | 2 | 3 | 4 | 5 | 6 | 7
2 | 3 | 4 | 5 | 6 | 7 | 8
3 | 4 | 5 | 6 | 7 | 8 | 9 | cell position = (row die, column die)
|-----|-----|-----|-----|-----|-----| printed value = the sum; each cell = 1/36
4 | 5 | 6 | 7 | 8 | 9 | 10
5 | 6 | 7 | 8 | 9 | 10 | 11
6 | 7 | 8 | 9 | 10 | 11 | 12
^

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die 1

count the 7s: 6 of them $\rightarrow P(\text{sum}=7) = 6/36 = 1/6$ (the most likely sum)

count the 2s: 1 (top-left) $\rightarrow 1/36$

count the 12s: 1 (bottom-right) $\rightarrow 1/36$

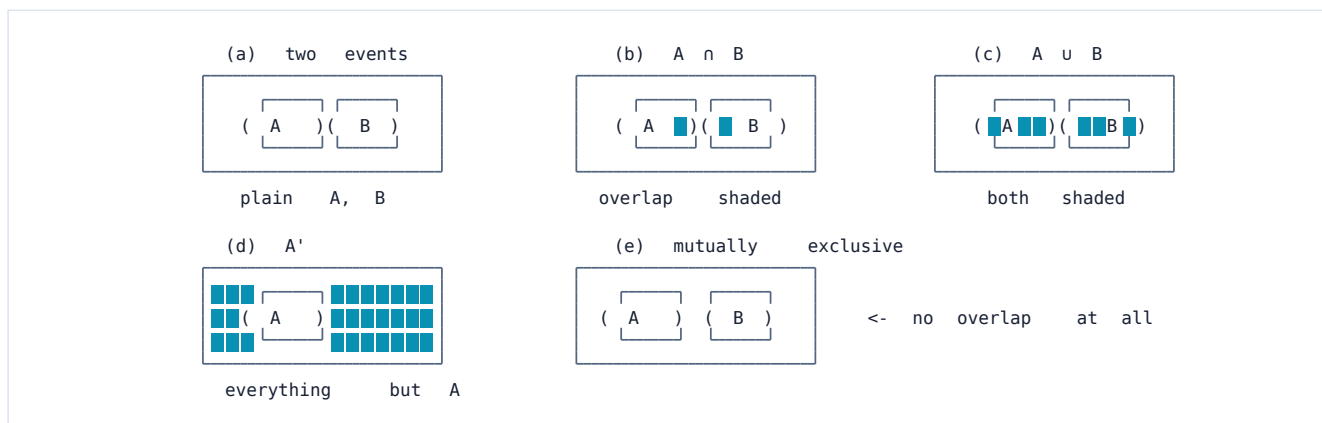
the counts run 1,2,3,4,5,6,5,4,3,2,1 (sums 2..12) = the familiar triangle

5. Events and set algebra (p030-p040)

outcome	a single possible result
event	a SUBSET of S (a collection of outcomes)
null event	the empty set, no outcomes

A' complement "not A" everything in S not in A
 $A \cup B$ union "A or B" in A, or B, or BOTH
 $A \cap B$ intersection "A and B" in BOTH
disjoint / mutually exclusive : $A \cap B = \emptyset$

ASCII Venn diagrams:



Demorgan, drawn as the two rules the slides state:

$$(A \cup B)' = A' \cap B' \quad (A \cap B)' = A' \cup B'$$

"neither" = not-A and not-B "not both" = not-A or not-B

6. Mutually exclusive vs independent (p025-p028) - THE classic trap

MUTUALLY EXCLUSIVE	INDEPENDENT
cannot happen together	one happening does not change the chance of the other
$A \cap B = \emptyset$	$P(A \cap B) = P(A) \times P(B)$
$P(A \cup B) = P(A) + P(B)$	or $P(A B) = P(A)$

THEY ARE NOT THE SAME, AND THEY ARE NOT OPPOSITES
exclusive $\rightarrow$ CANNOT both happen $\rightarrow$ knowing A KILLS B
(so NOT independent, when $P > 0$)

Worked: the die counter-example (p025)

$A = \text{even} = \{2,4,6\}$	$B = \text{prime} = \{2,3,5\}$
$A \cap B = \{2\}$	$\rightarrow$ they SHARE the outcome 2
$\rightarrow$ NOT mutually exclusive	
$A = \text{even} = \{2,4,6\}$	$B = \text{odd} = \{1,3,5\}$
$A \cap B = \emptyset$	$\rightarrow$ mutually exclusive

Worked: coin + die independence (p027)

A = head,	$P(A) = 1/2$
B = 6,	$P(B) = 1/6$
$P(A \cap B) = 1/2 \times 1/6 = 1/12$	-> independent

Worked: draw + replace (p027) -> independent. draw, no replace -> dependent.

Mutual independence of n events (p028): for EVERY subset of size k ($k = 2..n$),

$$P(A_{i1} \cap A_{i2} \cap \dots \cap A_{ik}) = P(A_{i1}) P(A_{i2}) \dots P(A_{ik})$$

7. Counting: the engine of half the course (p043)

ORDER MATTERS ?	ORDER DOES NOT MATTER ?
PERMUTATION $P(n, r) = n! / (n-r)!$ "arrangements"	COMBINATION $C(n, r) = n! / (r!(n-r)!)$ "choose", "committee", "sets of"
with REPEATED items: $n! / (n_1! n_2! \dots n_k!)$	$n_1, n_2, \dots$ are the repeat counts

WORD TRIGGERS

"arrange", "lineup", "order", "rank" -> PERMUTATION

"choose", "select", "committee", "set" -> COMBINATION

and $C(n, r) = C(n, n-r)$ (choosing r to keep = choosing n-r to drop)

8. The axioms and rules (p044-p045)

AXIOMS	
$P(S) = 1$	
$P(A) \geq 0$	
disjoint $A_1, A_2, \dots$:	$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$
RULES	
addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
disjoint case	$P(A \cup B) = P(A) + P(B)$
multiplication $P(A \cap B) = P(A) P(B A) = P(B) P(A B)$	
independent	$P(A \cap B) = P(A) P(B)$
complement	$P(A') = 1 - P(A)$

The addition-rule picture (why you subtract the overlap):

A $A \cap B$ B	$P(A) + P(B)$ counts $A \cap B$ TWICE
counted	subtract it ONCE
twice	

9. Worked example 1: tournament (p041-p042)

Four universities 1,2,3,4. Round 1: 1v2, 3v4. Winners play for the title, losers play too.

one outcome is written as a 4-letter code, e.g. 1324:

1 beats 2, 3 beats 4 (round 1) then 1 beats 3, 2 beats 4

$S = \{ 1324, 1342, 1423, 1432, 2314, 2341, 2413, 2431, 3124, 3142, 4123, 4132, 3214, 3241, 4213, 4231 \}$ -> 16 outcomes

(b) A = 1 wins the tournament	-> 1 must be FIRST in the code
$A = \{1324, 1342, 1423, 1432\}$	4 outcomes
(c) B = 2 gets into the final	-> 2 must be 1st or 2nd
$B = \{2314, 2341, 2413, 2431, 3214, 3241, 4213, 4231\}$	8 outcomes
(d) $A \cup B = 12$ outcomes	
$A \cap B = \emptyset$	because 1 and 2 CANNOT both reach the final
A'	$= S - A = 12$ outcomes

10. Worked example 2: faculty laptops (p046-p047)

6 faculty, 2 chose laptop, 4 chose desktop. Only 2 setups per day, the 2 are chosen randomly.

total outcomes = $C(6,2) = 15$ (choose 2 of the 6 computers)

(a) both laptops = $C(2,2)/15 = 1/15$

(b) both desktops = $C(4,2)/15 = 6/15 = 2/5$

(c) at least one desktop : complement is "both laptops" = A

$P(C) = 1 - P(A) = 1 - 1/15 = 14/15$

(d) at least one of each type:

$1 - P(A \cup B) = 1 - 1/15 - 2/5 = 1 - 1/15 - 6/15 = 8/15$

(A and B are disjoint, so no overlap to subtract)

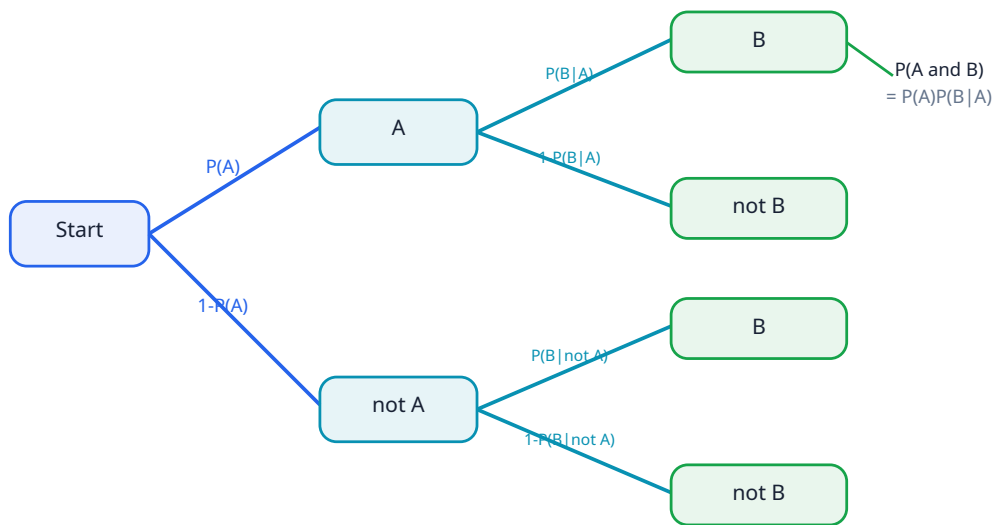
11. Worked example 3: basketball lineup (p048-p049)

STARTING LINEUP	=	2 guards	+	2 forwards	+	1 center	
Part (a):	3 centers,	4 guards,	4 forwards,	1 swing player	X		
no X		$C(4,2)$	x	$C(4,2)$	x	$C(3,1)$	= 6 x 6 x 3 = 108
X as guard		$C(4,1)$	x	$C(4,2)$	x	$C(3,1)$	= 4 x 6 x 3 = 72
X as forward		$C(4,2)$	x	$C(4,1)$	x	$C(3,1)$	= 6 x 4 x 3 = 72
							TOTAL = 252 lineups
Part (b):	5 guards, choose 5	5 forwards, of the 15	3 centers, at random;	2 swing (X and Y)	P(valid lineup)?		
neither X nor Y		$C(5,2)C(5,2)C(3,1)$				= 300	
X only		$C(5,1)C(5,2)C(3,1)$		x2 cases		= 300	
Y only		same				= 300	
both X and Y		$C(5,1)C(5,1)C(3,1)$		+ 2	$C(5,2)C(3,1)$	= 135	
						TOTAL = 1035	
P =	1035 / $C(15,5)$					= 1035 / 3003	

THE METHOD: split into disjoint cases by WHO of the swing players is used, count each case, ADD. Never double count.
no X, no Y + X only + Y only + both X and Y = total

12. Conditional probability (p050)

Conditional probability tree



Conditional probability tree

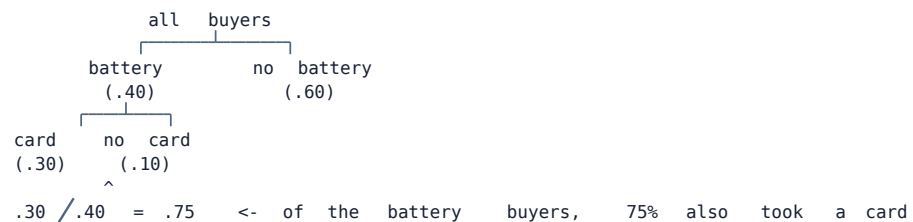
$$P(A|B) = P(A \cap B) / P(B), P(B) > 0$$

Worked: digital camera buyers

60% buy a memory card	$P(A) = .60$
40% buy an extra battery	$P(B) = .40$
30% buy both	$P(A \cap B) = .30$
$P(\text{card} \text{battery}) = .30 / .40 = .75$	
$P(\text{battery} \text{card}) = .30 / .60 = .50$	

NOTE THE TWO ANSWERS ARE DIFFERENT. .75 is NOT .50.
The denominator is WHAT YOU ARE CONDITIONING ON.
 $P(A|B)$ and $P(B|A)$ are different numbers.

The tree that makes it obvious:



That is lecture 2 in full. The two things to carry forward: (1) the 36-cell dice grid, (2) exclusive is not independent.